

Alex Welcing

Lupine Science

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github.com/alexwelcing/lupine

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Dear Editor,

I am pleased to submit our manuscript, “Hyper-Ribbon Manifolds of Prediction Error: Universal Low-Dimensional Structure in Parameterized Models, with Ecological Fallacy Detection,” for consideration as a Research Article in your journal.

This paper reports a discovery with relevance across physics, mathematics, statistics, and machine learning. By analyzing 559 independently parameterized models for predicting material properties, we demonstrate that prediction errors universally collapse onto low-dimensional hyper-ribbon manifolds — the signature of sloppy model universality. We further show that standard cross-system benchmarking commits ecological fallacy (Simpson's paradox) with extreme heterogeneity ($R^2 = 98.6\%$), and that this hyper-ribbon structure extends to foundation neural network potentials, establishing paradigm independence.

The work synthesizes information geometry, random-effects meta-analysis, and causal inference in a way we believe will interest readers from multiple disciplines. The framing is deliberately general: while the domain is interatomic potentials, the questions (geometric error structure, heterogeneity in multi-model benchmarks, confounder control) are universal.

No preprint has been published. All code and data are open-source (MIT License) at github.com/alexwelcing/lupine. The analysis engine, *atlas-distill*, is a standalone Rust binary.

Suggested reviewers:

Prof. James P. Sethna (Cornell University) — Pioneer of sloppy model theory; expertise in information geometry and model manifold dimensionality.

Prof. Judea Pearl (UCLA) — Founder of causal inference; expertise in Simpson's paradox, ecological fallacy, and confounder analysis.

Prof. Gábor Csányi (University of Cambridge) — Leading MLIP researcher (GAP, MACE); expertise in benchmarking and uncertainty quantification for materials models.

Sincerely,

Alex Welcing

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